Customer Segmentation for a Retail Brand

Module 4 Mini Project | K-means Clustering on RFM Features

1. Executive Summary

This project applies K-means clustering to customer-level Recency, Frequency and Monetary (RFM) features from the Online Retail transaction dataset. After cleaning, the analysis retained **397,884 valid transaction rows** and built RFM profiles for **4,338 customers**. Four clusters were fitted after log transformation and standardisation. The silhouette score was **0.337**, suggesting some useful structure but also overlap between groups. The segments support differentiated retention, cross-selling, VIP and win-back campaigns.

2. Business Problem and Objectives

Customers have different purchase patterns and spending levels. Sending the same promotion to everyone can waste marketing budget or discount customers who would buy anyway. Objectives: create customer groups from purchase history, profile groups using RFM, and recommend targeted marketing actions.

3. Dataset and Preparation

The dataset includes invoice-level transactions with invoice number, quantity, invoice date, unit price, customer ID and country. Rows without CustomerID or a valid date were removed. Cancelled invoices (InvoiceNo beginning with C), non-positive quantities and non-positive unit prices were excluded. Sales was calculated as Quantity × UnitPrice. Recency uses a snapshot date one day after the latest transaction: **10 December 2011**.

4. Methodology

Recency: days since the customer's latest purchase (lower is more recent).
Frequency: distinct invoices per customer.
Monetary: total transaction sales per customer.
Transformation: log(1 + feature) reduces skew.
Scaling: StandardScaler puts features on comparable scales.
Clustering: K-means with k = 4, random_state = 42 and n_init = 10.

5. Results and Segment Profiles

Cluster IDs are arbitrary labels, not ranks. Interpret them using the RFM profiles below.

Cluster	Customers	Customer share	Median recency	Median orders	Median spend	Revenue share
0	1,622	37.4%	173 days	1	£296.82	6.2%
1	810	18.7%	16 days	2	£470.12	5.0%
2	713	16.4%	8 days	10	£3,729.34	64.7%
3	1,193	27.5%	54 days	4	£1,341.65	24.1%

6. Targeting Strategy

Cluster 0 — At-risk / dormant, low value
High recency and typically low frequency/spend. Use a low-cost win-back email, relevant product reminders and test a small time-limited incentive. Stop repeated offers to customers who do not respond.

Cluster 1 — Recent / occasional customers
Recent purchase but typically fewer orders and lower spend. Use a welcome/post-purchase journey, complementary product recommendations and a gentle second-purchase incentive.

Cluster 2 — Champions / VIP customers
Very recent, frequent purchases and high spend. Protect retention with loyalty perks, early access, personalised recommendations and referral rewards. Avoid blanket discounts.

Cluster 3 — Regular customers / growth potential
Repeat activity and meaningful spend, with room to grow. Use bundles, replenishment reminders and relevant cross-selling to increase basket size and order frequency.

7. Implementation Plan and Success Measures

Export customer IDs with cluster labels into the campaign platform. Test each segment's campaign with a holdout group where possible. Track repeat-purchase rate, conversion rate, revenue per contacted customer, average order value, unsubscribe rate and campaign cost. Review segment profiles monthly or quarterly.

8. Conclusion

RFM-based K-means provides an explainable starting point for customer segmentation. Prioritise retention and advocacy for high-value active customers, cross-selling for regular customers, and carefully measured win-back campaigns for dormant customers. The silhouette score indicates that segments are useful but not perfectly separated, so business validation and campaign experiments are important.

Limitations

The dataset is historical and ends in 2011, so it is for learning and should not be treated as a current customer profile. Monetary is sales, not profit. No demographic or marketing-response variables are included. Four clusters were selected for interpretability; other values of k may also be compared. Segment labels are human interpretations, not ground-truth classes.